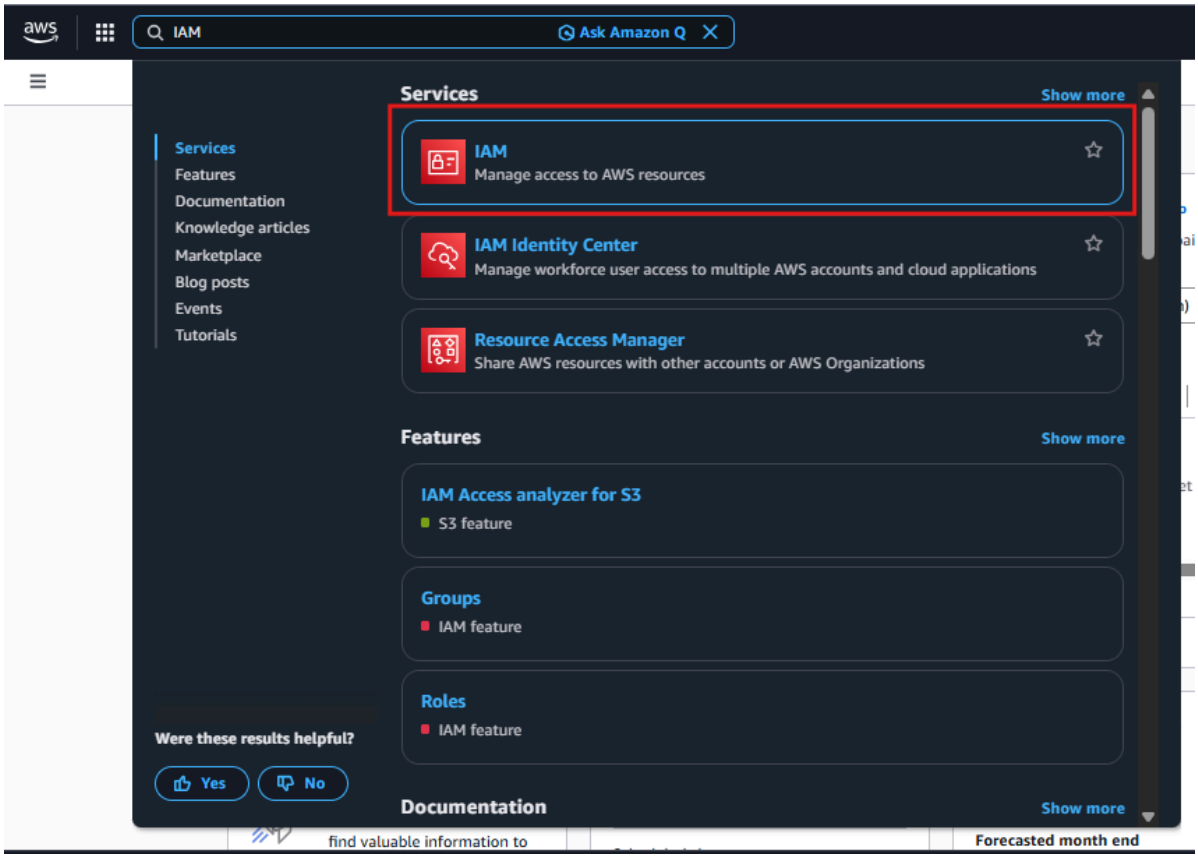
Lab: Deploy Spark and Python Jobs on AWS Glue

Objective

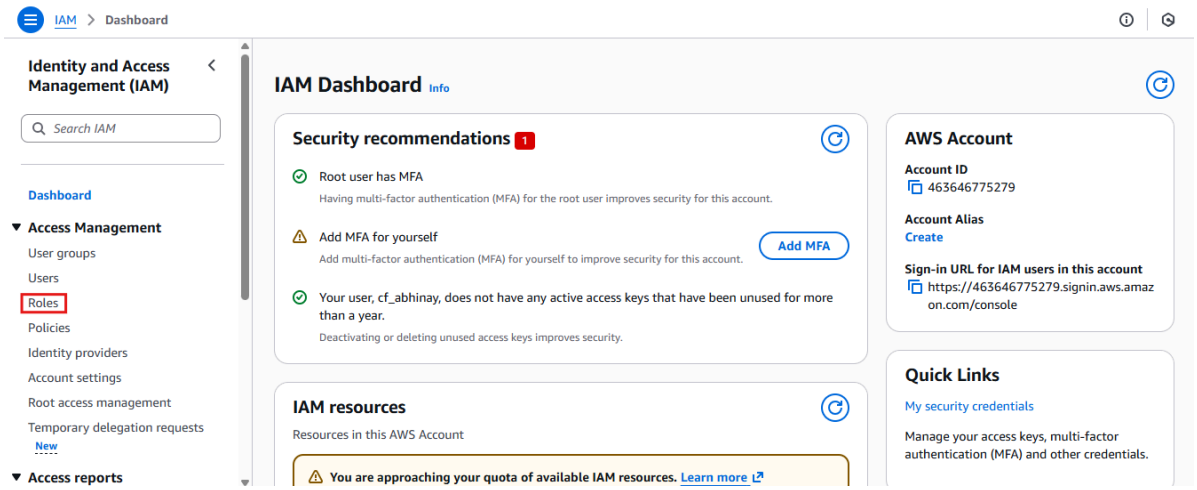
Deploy a locally tested Spark ETL script and a Python shell script to AWS Glue, configure required IAM permissions, and prepare both jobs for orchestration via Airflow.

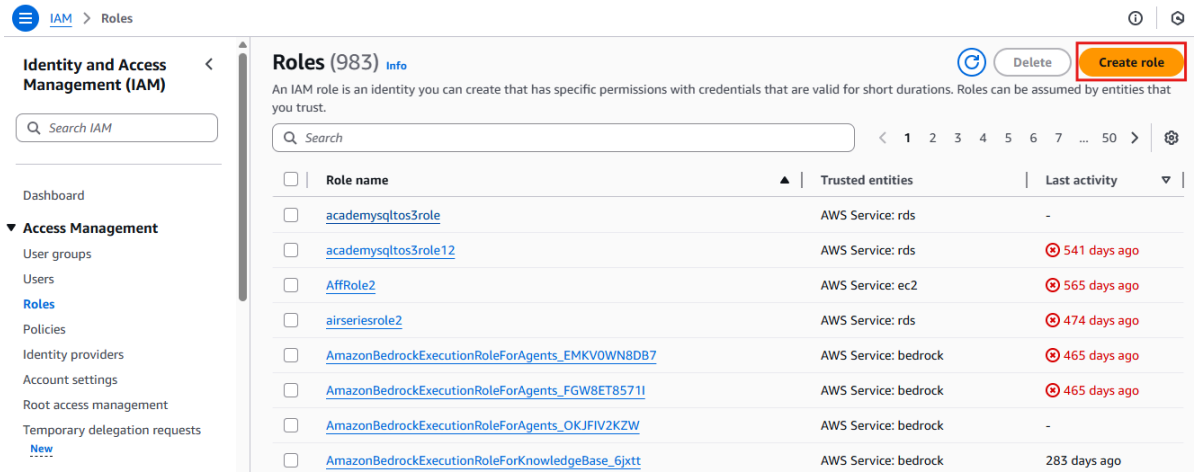
Step 1: Create IAM Role for AWS Glue Jobs

1. Open the AWS Console and navigate to **Identity and Access Management (IAM)**.

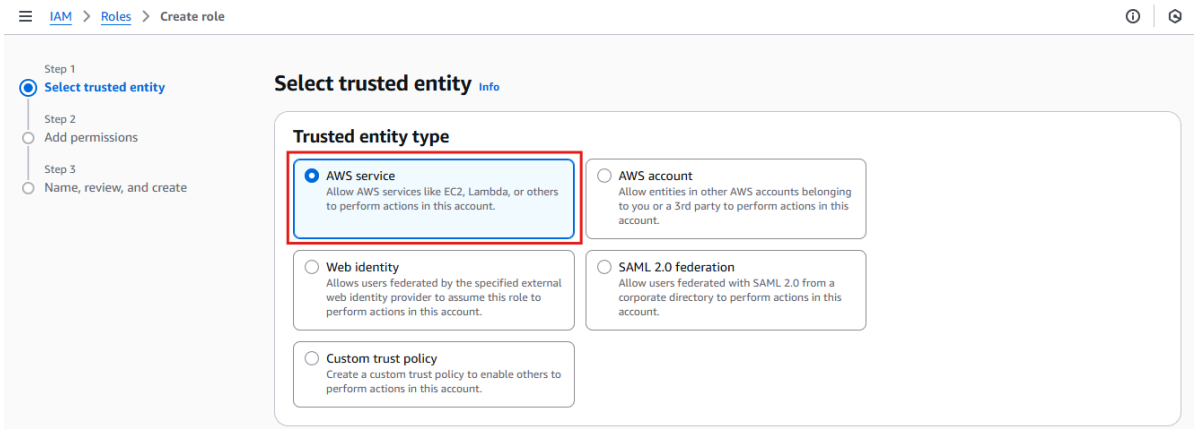


2. Select **Roles** and click **Create role**.

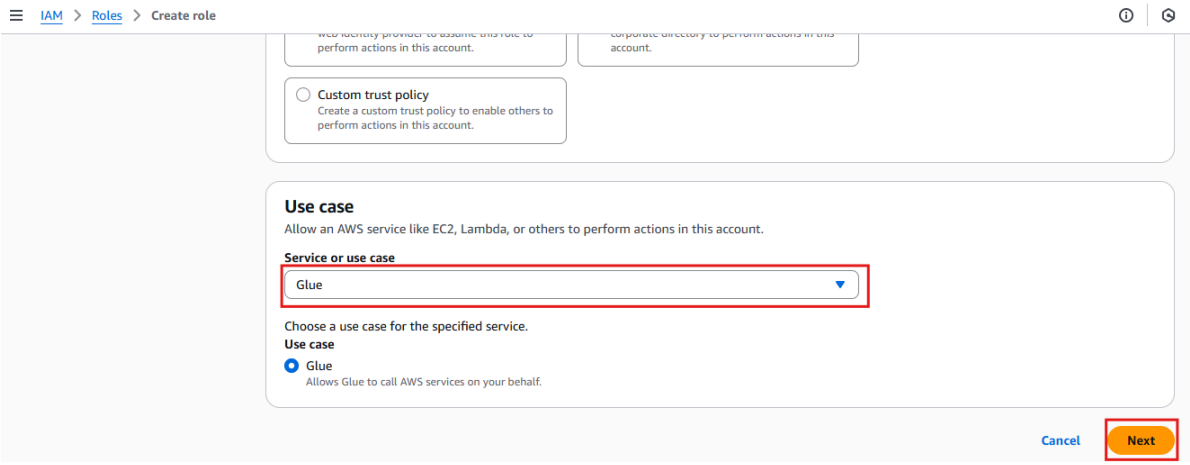




3. Choose **AWS service** as the trusted entity type.

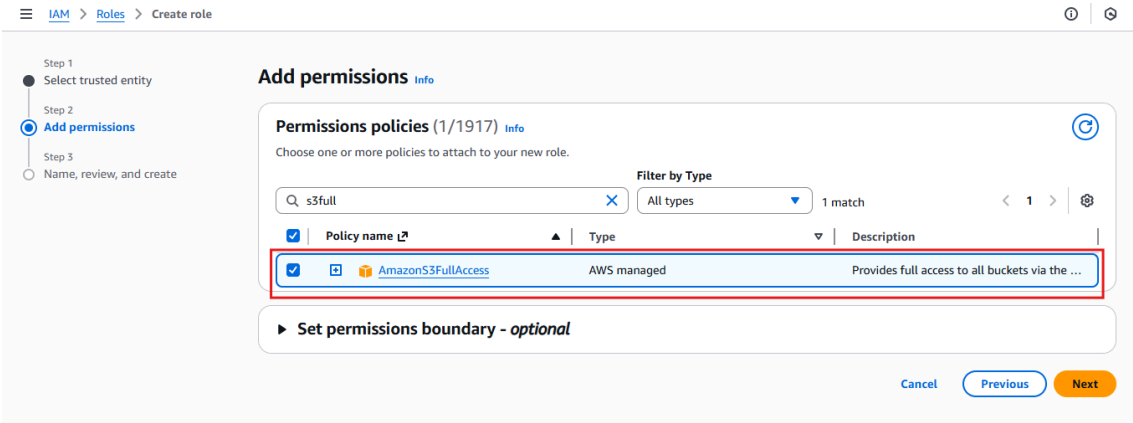


4. From the service list, select **Glue** and proceed to the next step.

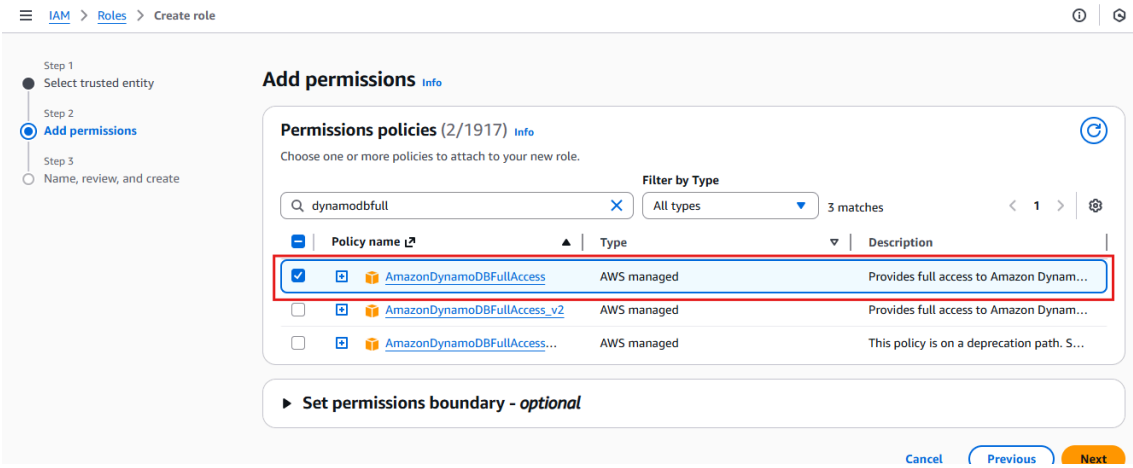


5. Attach the following permission policies to the role:

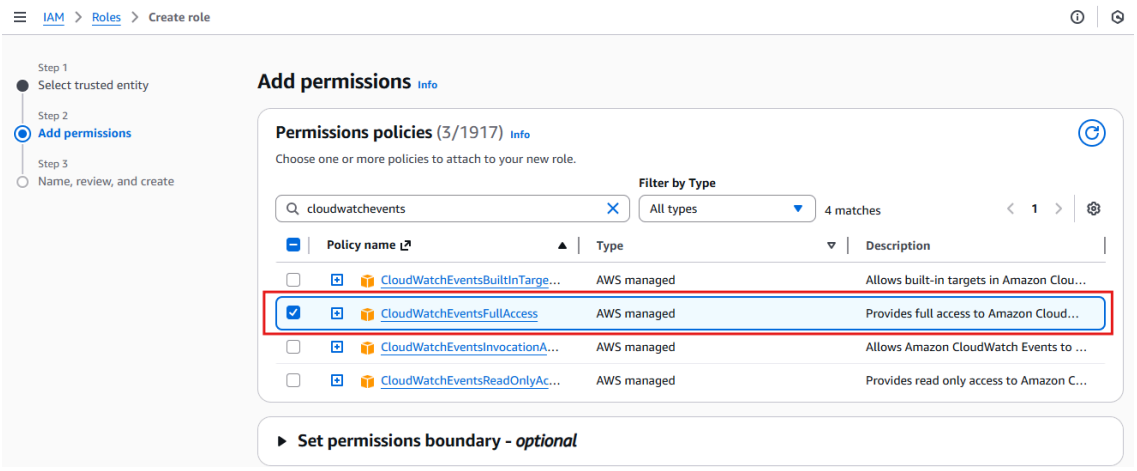
- **AmazonS3FullAccess**



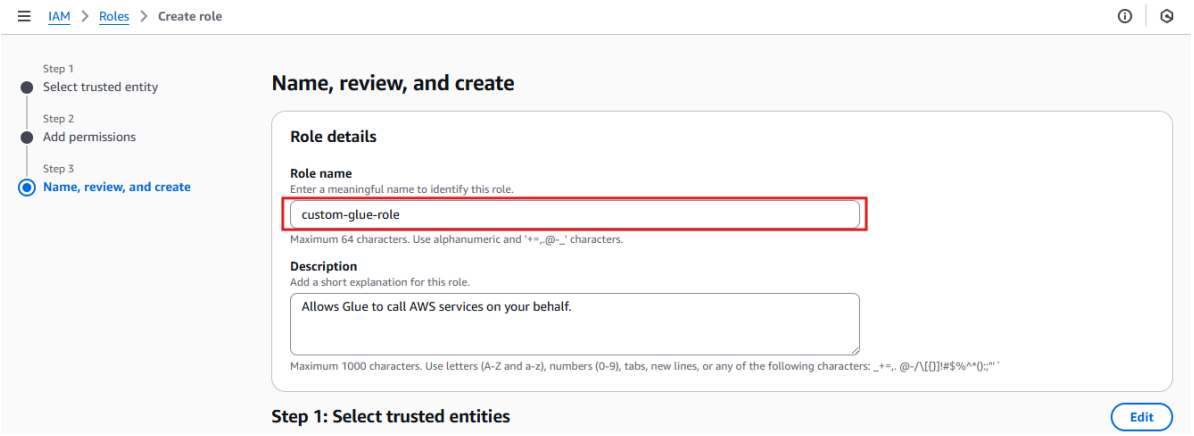
- **AmazonDynamoDBFullAccess**

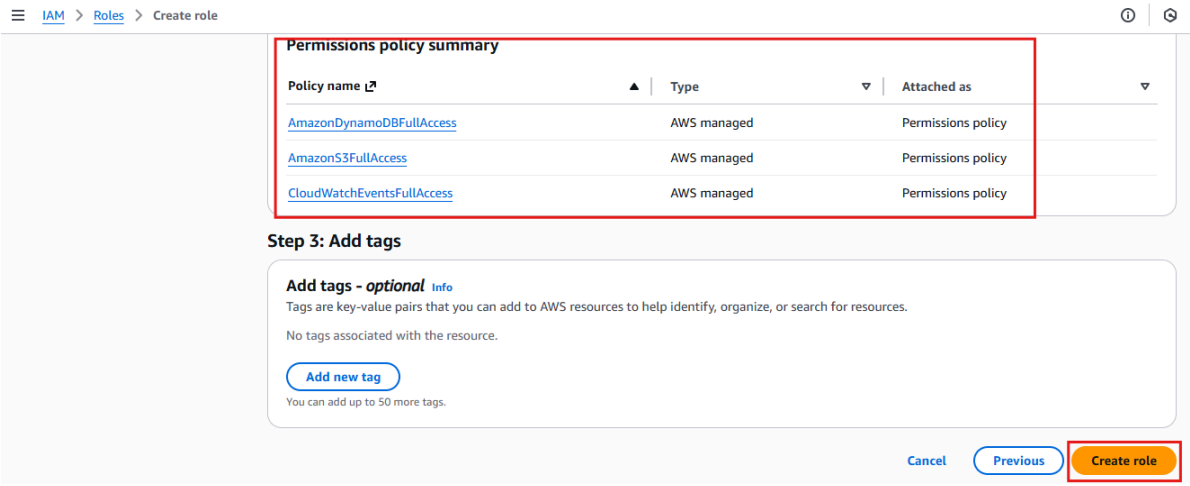


- **CloudWatchEventsFullAccess**



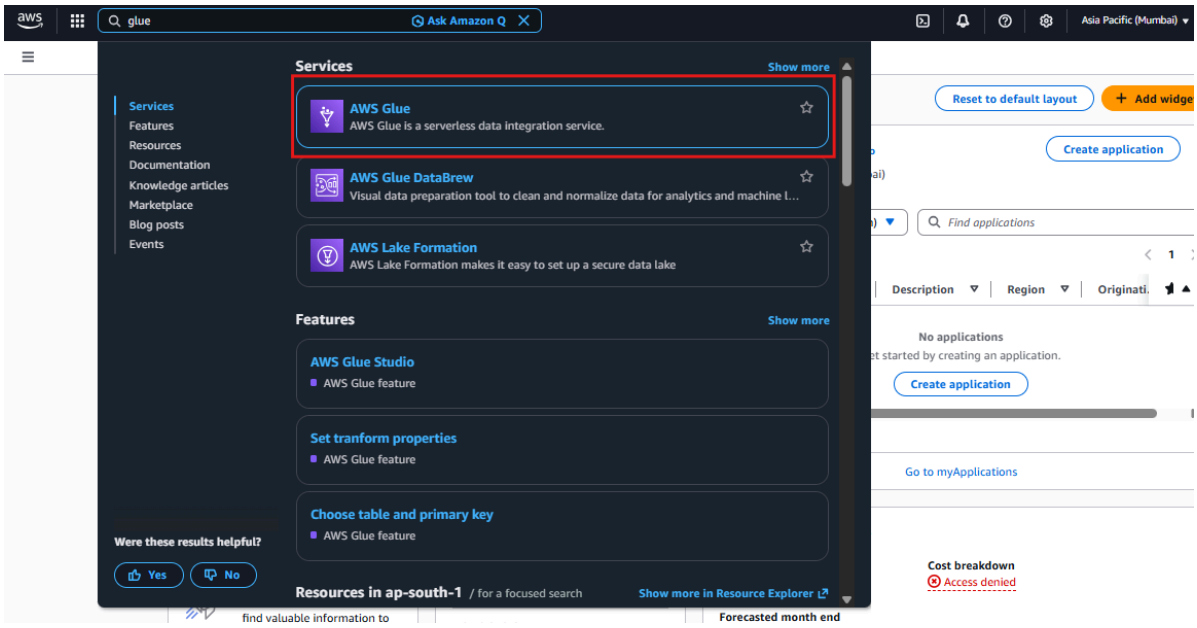
6. Ensure all three policies are selected, as they are required for:
   - Reading and writing data to S3
   - Performing upsert operations in DynamoDB
   - Writing logs to Amazon CloudWatch
7. Click **Next**, provide a meaningful role name (for example, `CustomGlueRole`), review the attached policies, and create the role.



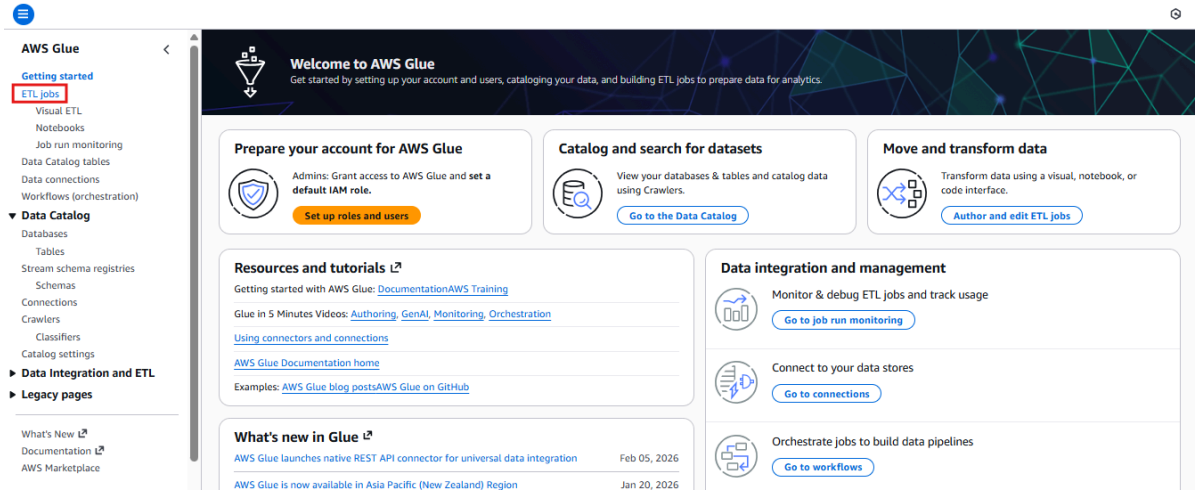


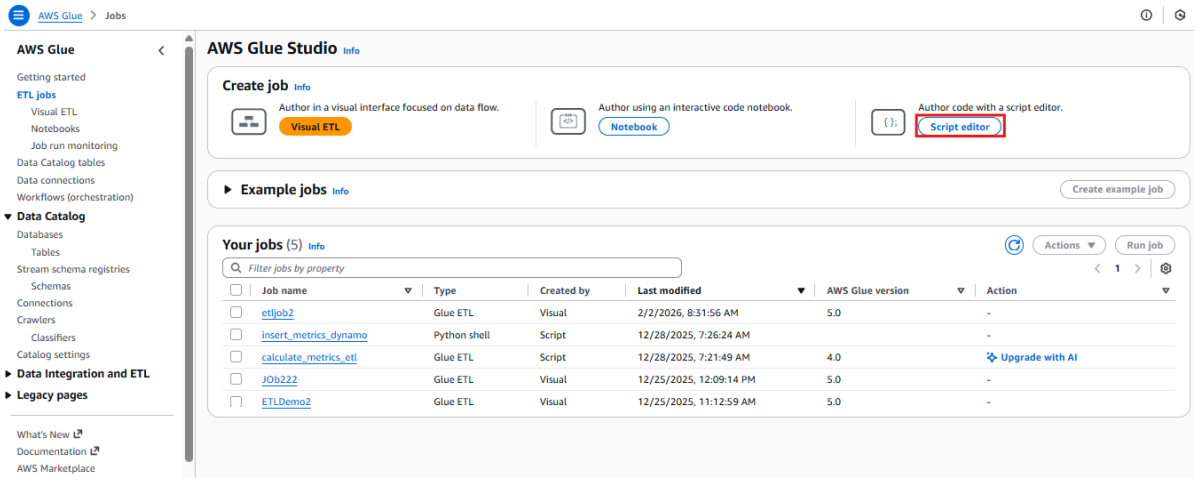
Step 2: Deploy the Spark ETL Job in AWS Glue

1. Navigate to **AWS Glue** from the AWS Console.

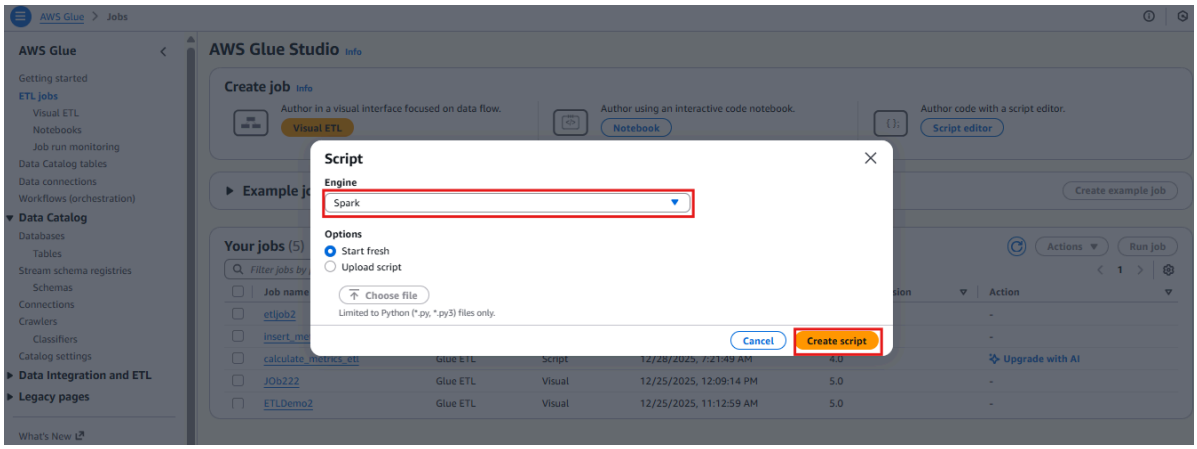


2. Select **ETL jobs** and click **Script editor**.

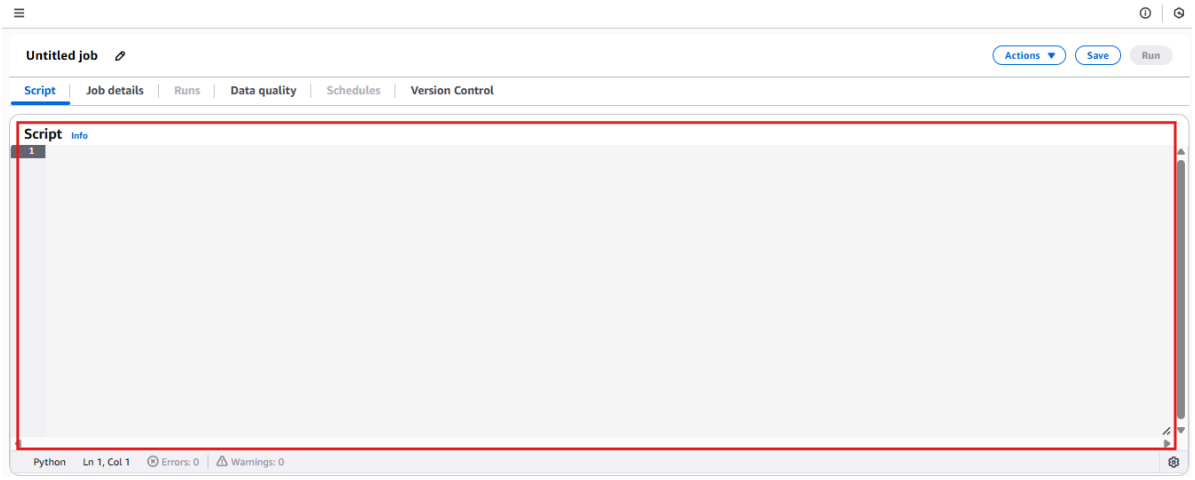


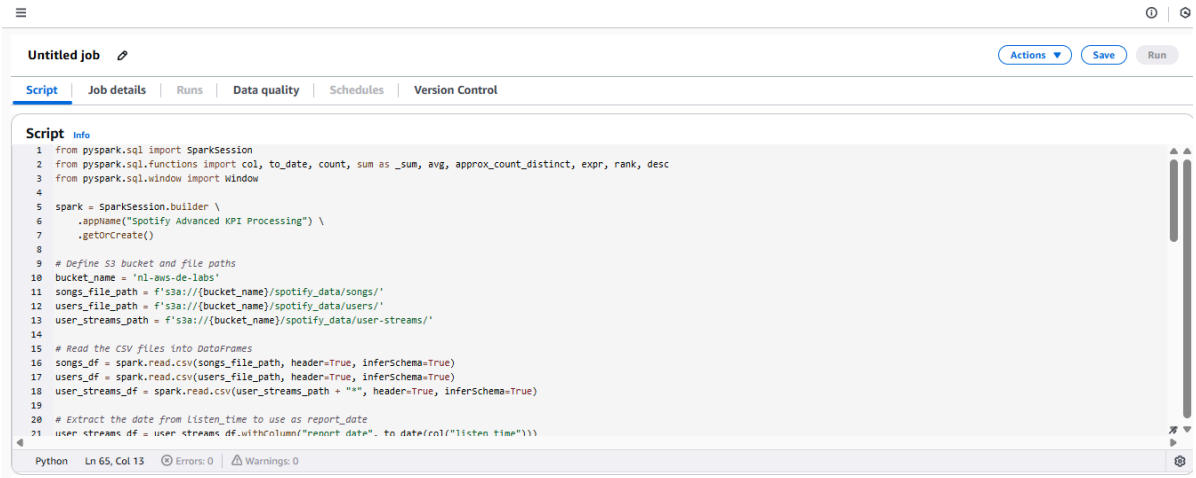


3. Choose **Spark** as the engine and click **Create script**.

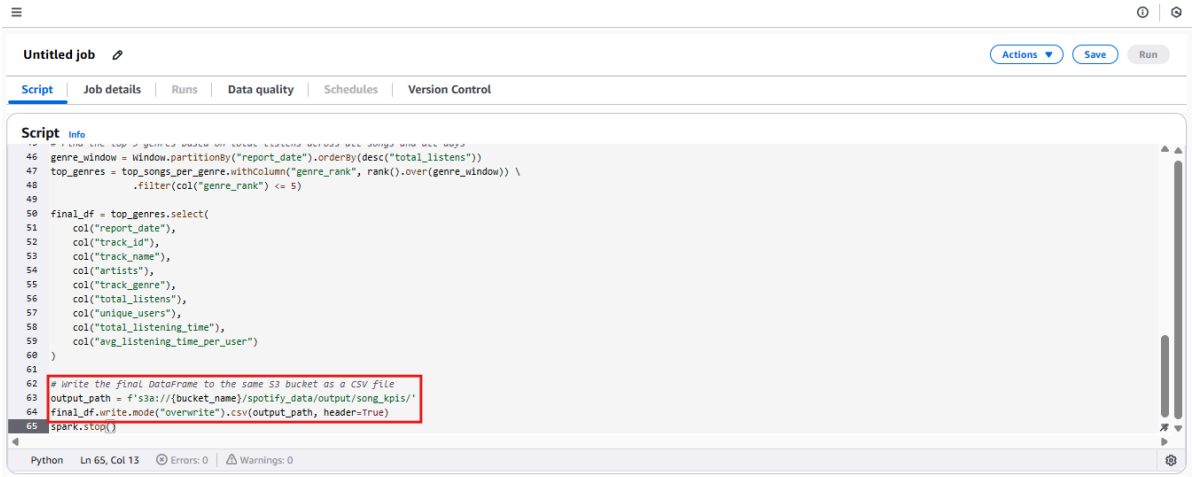


4. Copy the contents of your local PySpark script and paste it into the editor.



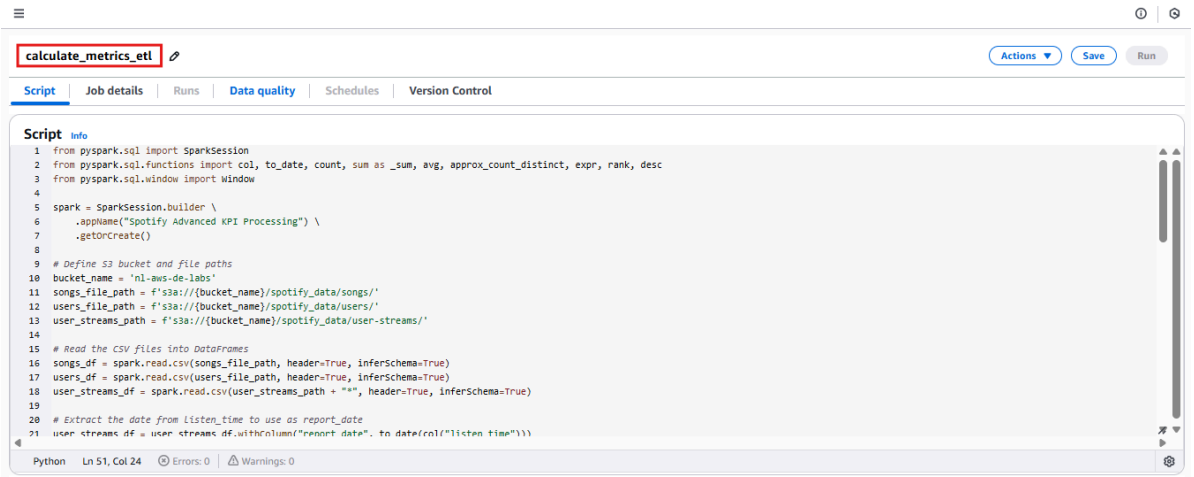


5. Ensure the final action in the Spark code writes the output to S3.

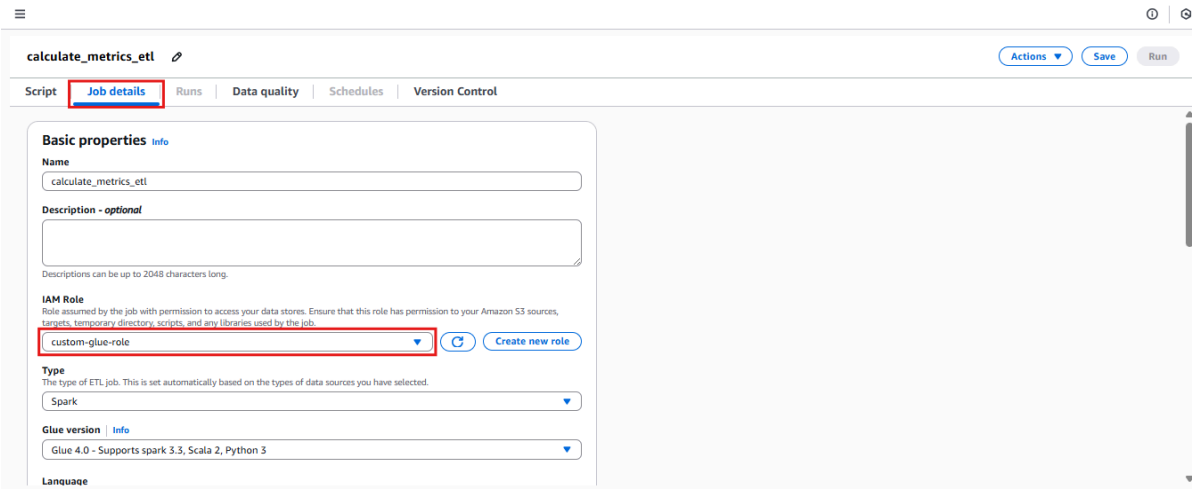


Note: Comment out any `df.show()` statements and enable the line that writes the DataFrame to S3.

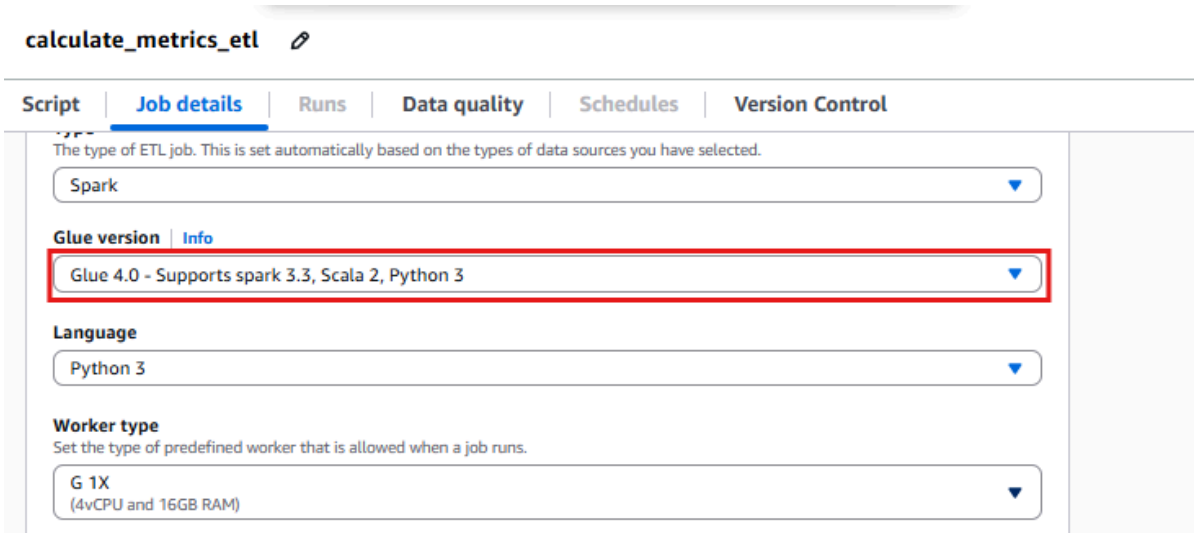
6. Set the job name to **Calculate Metrics ETL**.



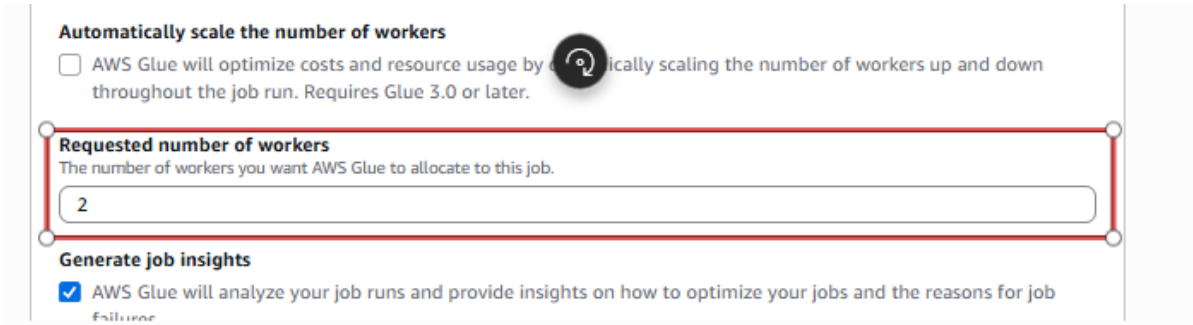
7. In **Job details**, select the IAM role created earlier.



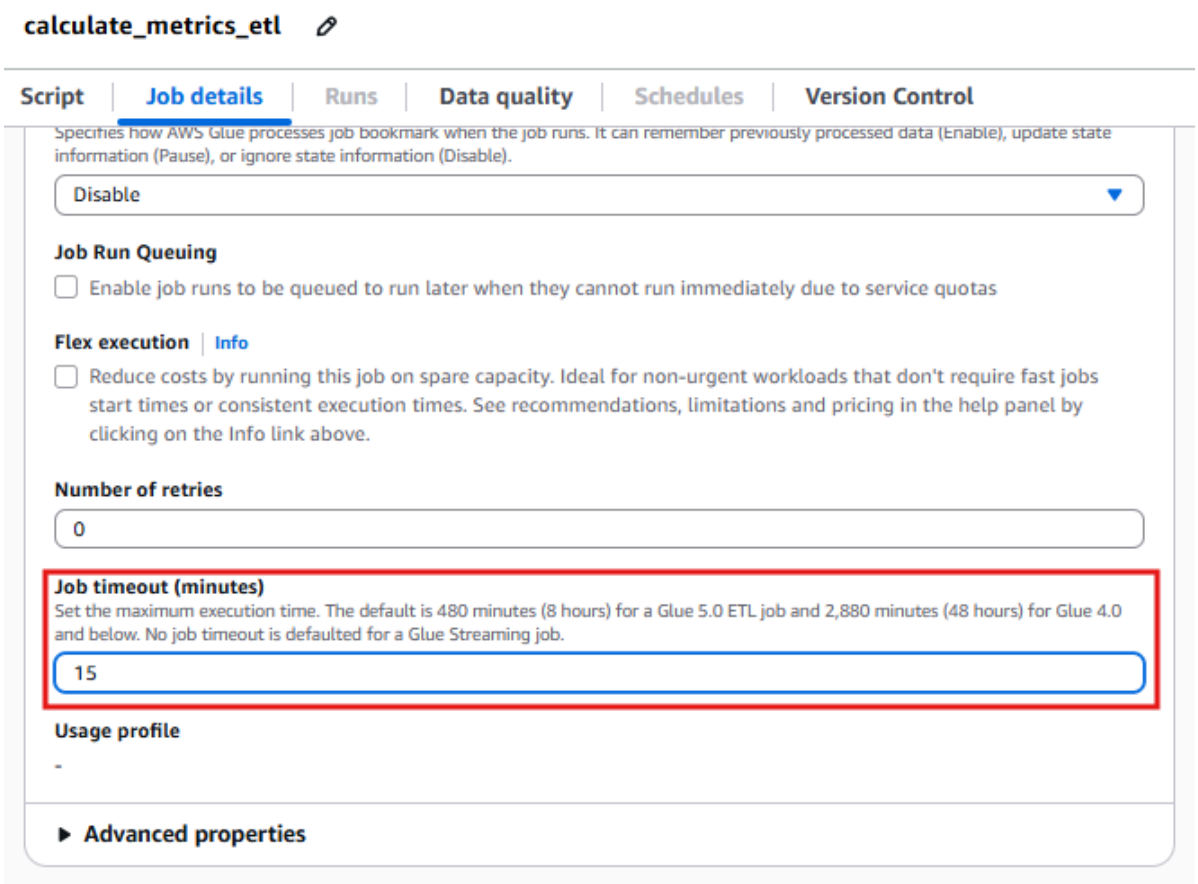
8. Set the Glue version to **Glue 4.0**.



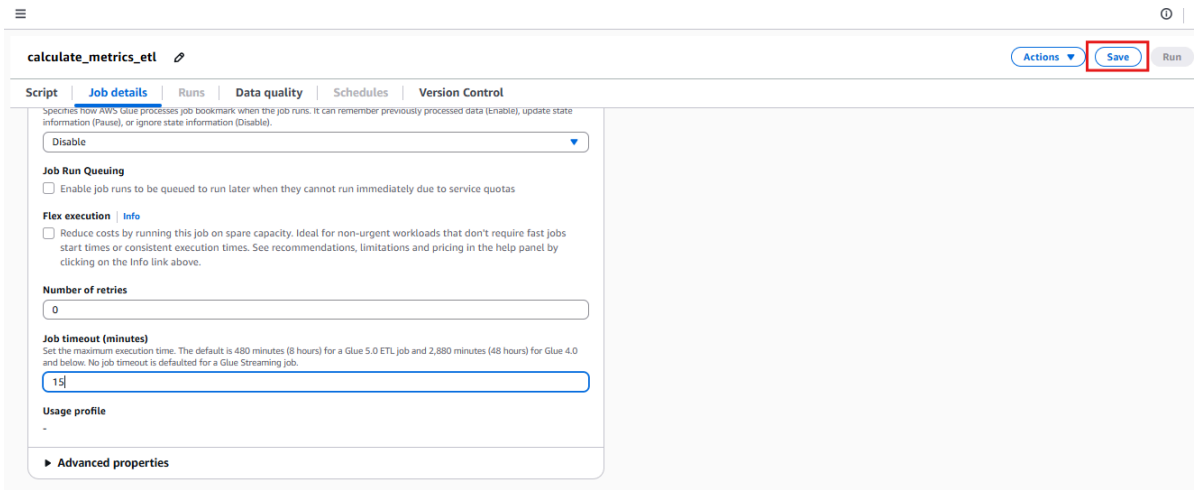
9. Configure the number of workers to **2**.



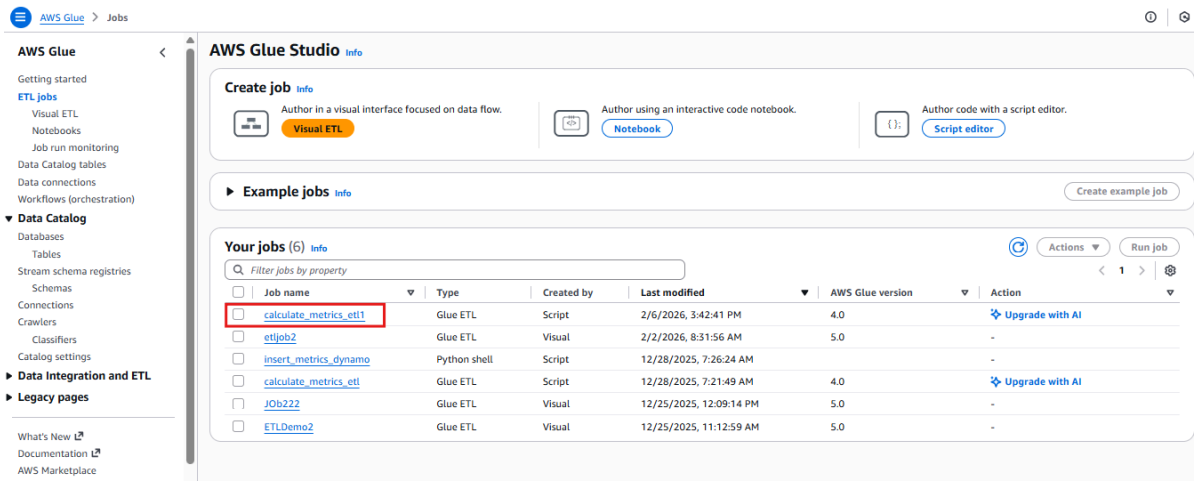
10. Set the job timeout to **15 minutes**.



11. Save the job.

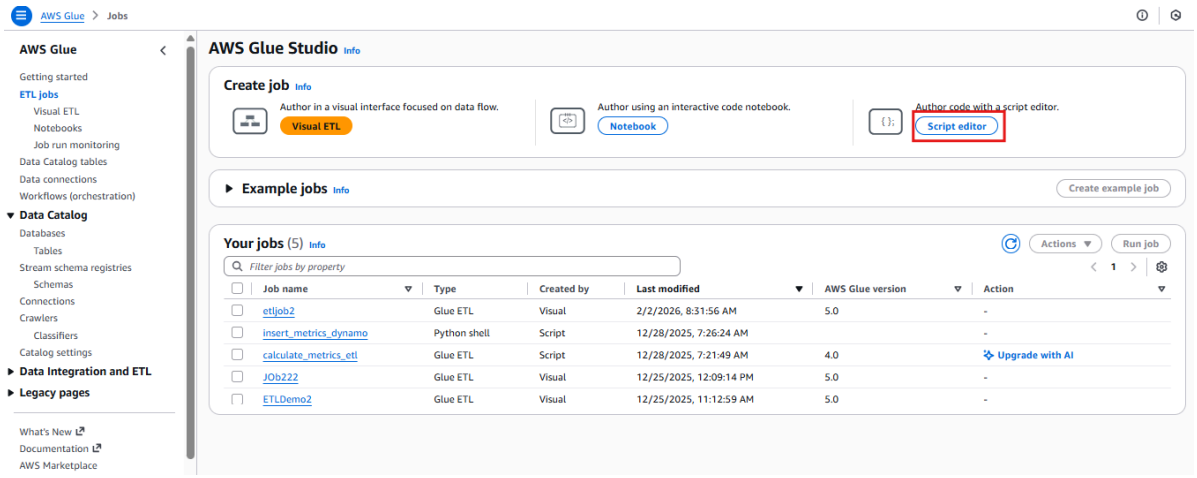


12. Return to the ETL jobs list and refresh to confirm the Spark job is created.

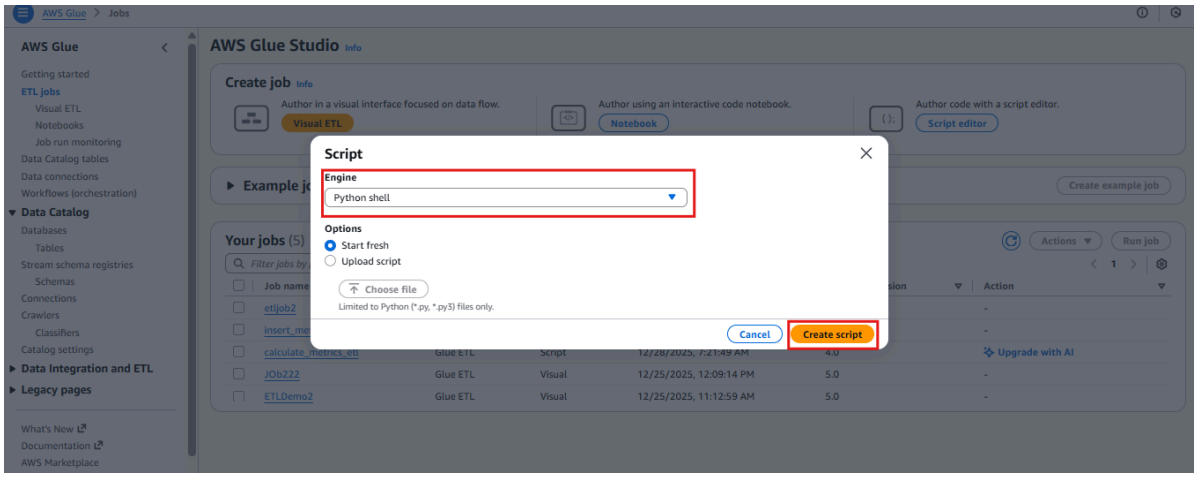


Step 3: Deploy the Python Shell Job for DynamoDB Upserts

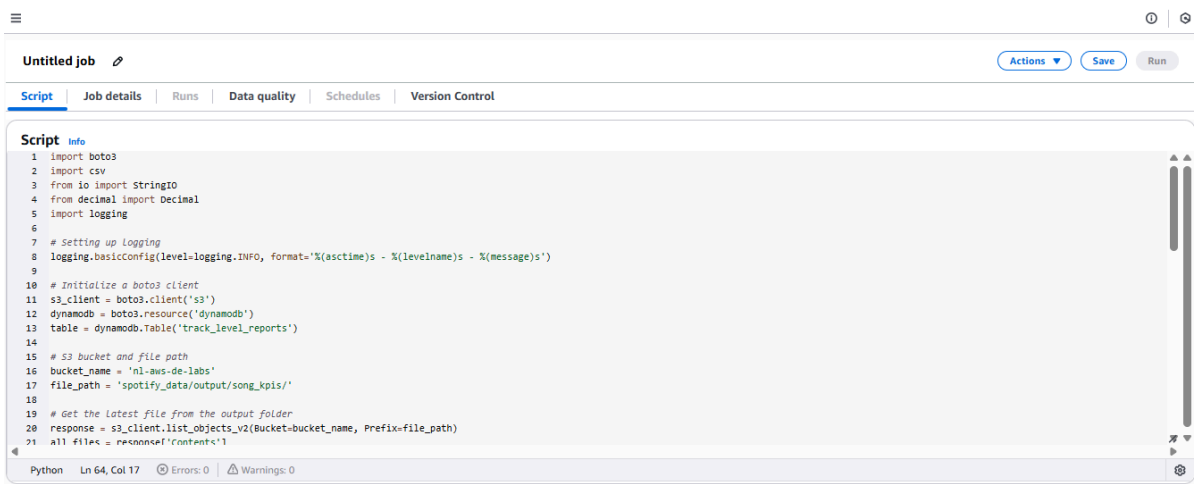
1. In AWS Glue, click **Script editor** again.



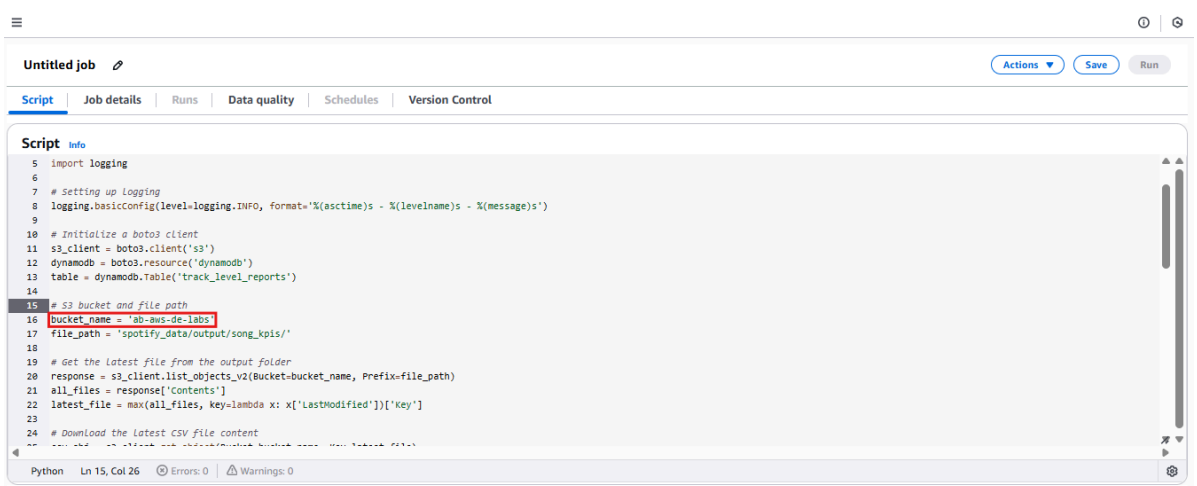
2. Select **Python Shell** as the engine.



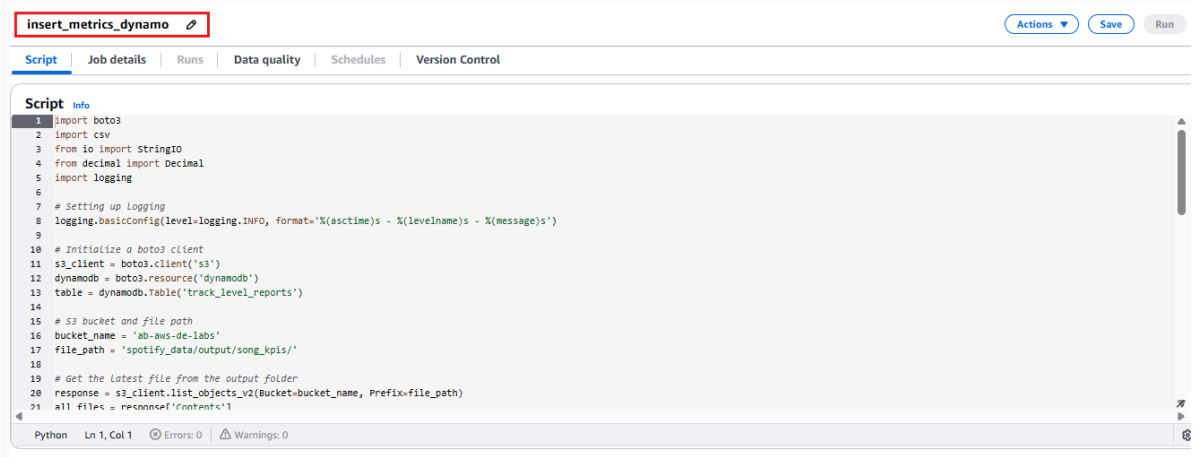
3. Paste the Python script into the editor.



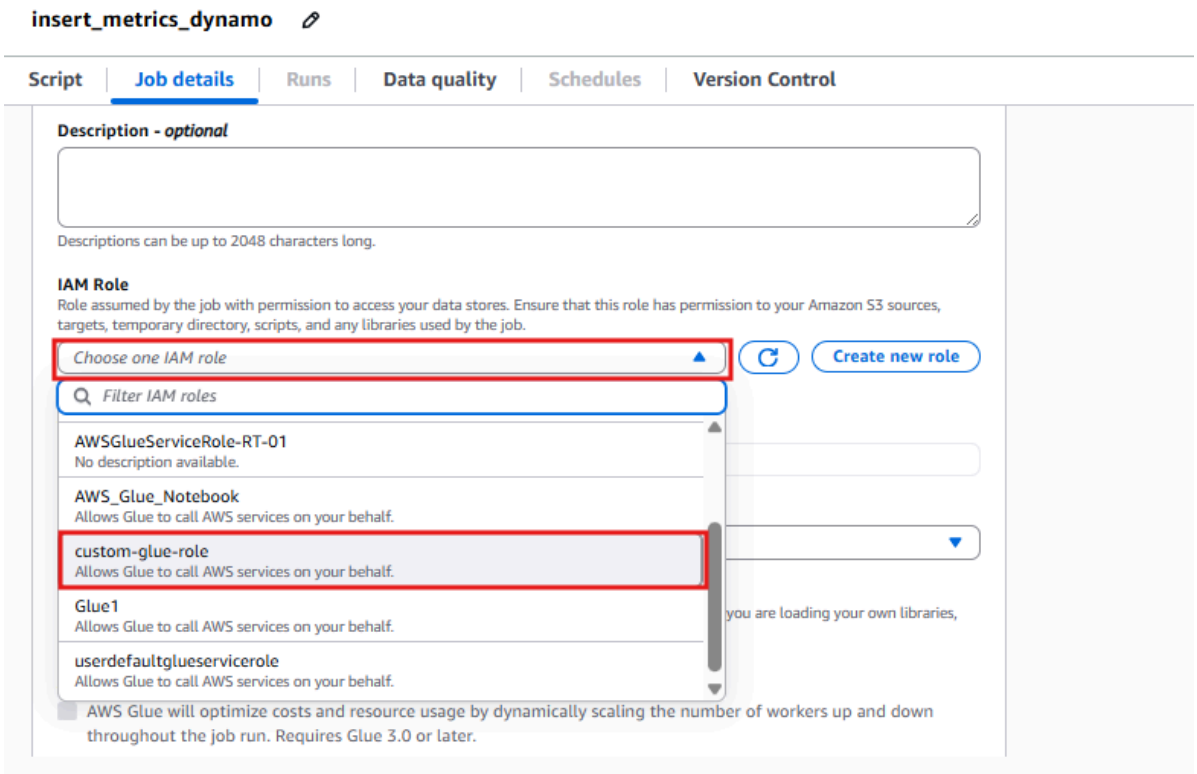
4. Update the S3 bucket name in the script if required.



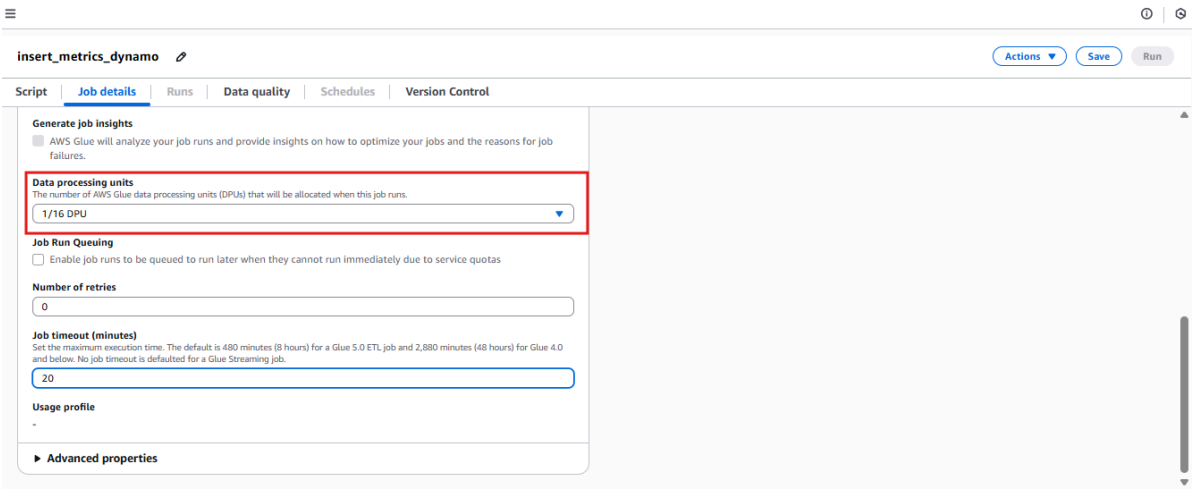
5. Set the job name to **Insert Metrics Dynamo**.



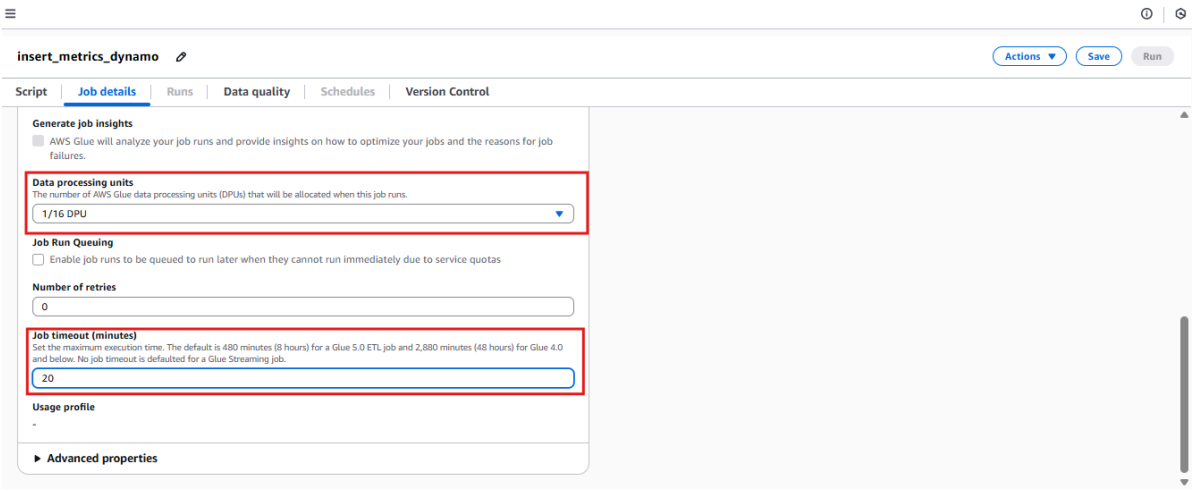
6. Under **Job details**, select the same IAM role used for the Spark job.



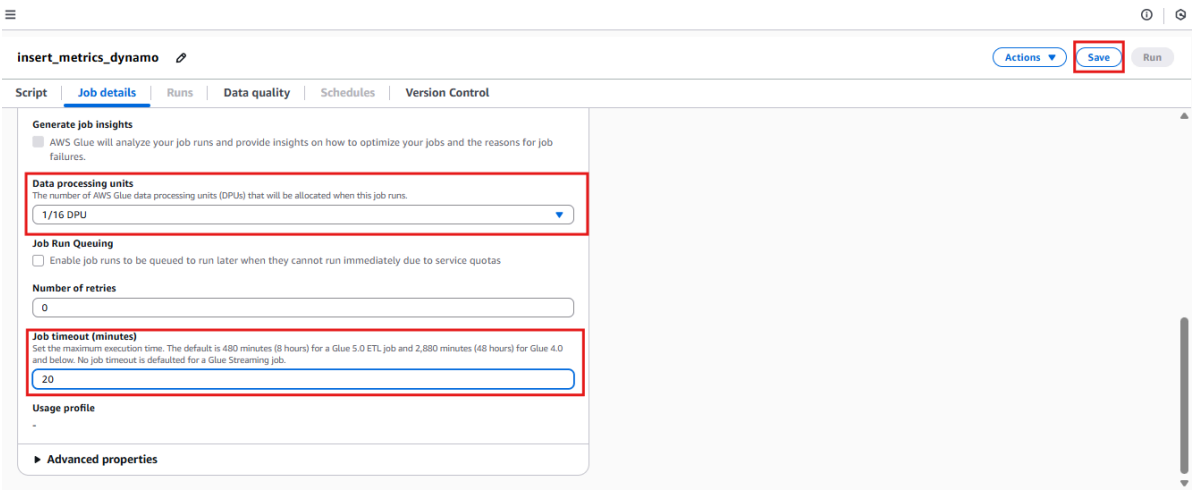
7. Leave the DPU setting at **1/16 DPU**.



8. Set the timeout to **20 minutes**.

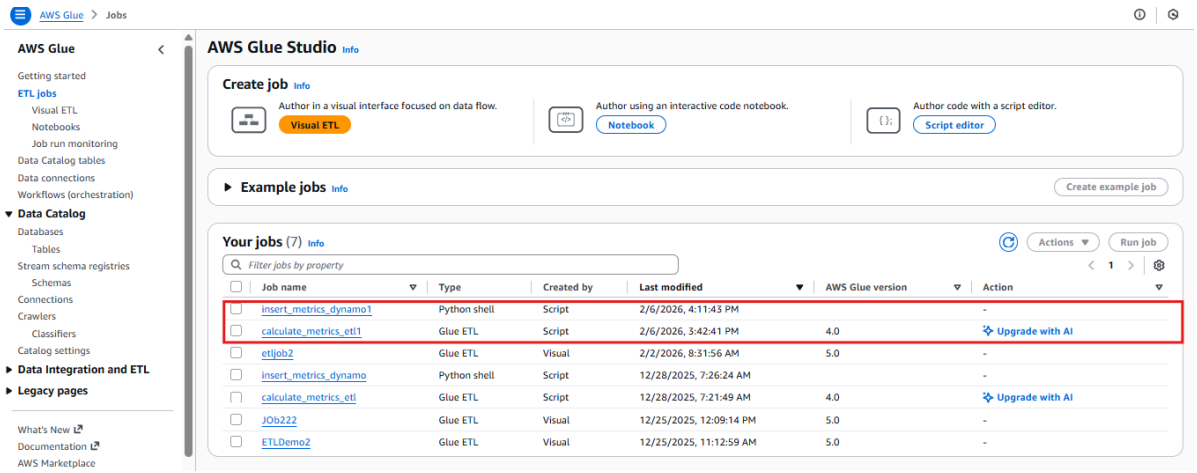


9. Save the job.



Step 4: Verify Glue Jobs

1. Navigate back to **ETL jobs** in AWS Glue.
2. Refresh the page.
3. Confirm the presence of two jobs:
    - **Calculate Metrics ETL** (Spark job)
    - **Insert Metrics Dynamo** (Python Shell job)



Outcome

Both Spark and Python Glue jobs are successfully deployed with the required IAM permissions and configurations. These jobs are now ready to be triggered manually or orchestrated via Airflow.